14/08/25
L-8

Final Exam

- → Laplace Transformation → (R.B.) → Differential Equations. (S.I.Ross) → Ch-9
- → Complex analysis (Ch-4, 5, 7)

Laplace transformation

② { → L.T. (Laplace transform).
→ Inverse Laplace Transform

② * → Application
→ Ordinary D.E.
→ System of O.D.E.

Laplace Transform

" Mathematical technique that changes a function of time into a function in the frequency domain."

Defn: $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t)\, dt$. → "s" is the complex number.

① $\boxed{\mathcal{L}\{1\} = \frac{1}{s}}$ $f(t) = 1$.

$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$

$\therefore \mathcal{L}\{1\} = \int_0^\infty e^{-st}(1)\, dt$

$= \int_0^\infty e^{-st}\, dt$

$= \lim_{a\to\infty} \int_0^a e^{-st}\, dt$

$= \lim_{a\to\infty} \left[\frac{e^{-st}}{-s}\right]_0^a$

* improper integral

i) $\int_0^\infty$, $\int_{-\infty}^\infty$

ii) $\int_0^5 \frac{dx}{x-3}$ → discontinue at a point at $x=3$, $\frac{1}{x-3}$ is undefined.

iii) combined

$\int_0^\infty \frac{dx}{x}$ → type (i) and (ii) combined.

$\boxed{\text{\#use limit to solve}}$

$\int e^{2x}\, dx = \frac{e^{2x}}{2}$

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$$= -\frac{1}{s} \lim_{a\to\infty} \left(e^{-sa} - e^{0}\right).$$

$$= -\frac{1}{s}\left(\lim_{a\to\infty} e^{-sa} - \lim_{a\to\infty}(1)\right)$$

$$= -\frac{1}{s}(0-1)$$

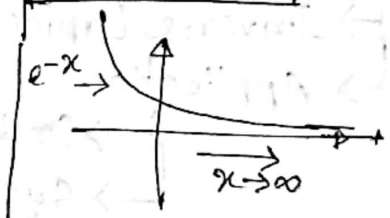
$$= \frac{1}{s}.$$

$$\therefore \boxed{\mathcal{L}\{1\} = \frac{1}{s}.}$$

$$\lim_{a\to\infty} e^{-as} = \lim_{a\to\infty} \frac{1}{e^{as}}$$

$$= \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$$

$$\boxed{\lim_{a\to\infty} e^{-as} = 0}$$



$$\lim_{x\to\infty} e^{-x} = 0$$

(2) $f(t) = \cos bt$; Find Laplace

$\mathcal{L}\{\cos bt\} = ?$

Sol^n: $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$

$$\Rightarrow \mathcal{L}\{\cos bt\} = \int_0^\infty e^{-st}(\cos bt)\,dt$$

$$\cos t = \frac{e^{it} + e^{-it}}{2}$$

$$= \int_0^\infty e^{-st}\,\frac{e^{ibt} + e^{-ibt}}{2}\,dt$$

$$= \frac{1}{2}\int_0^\infty \left(e^{-st}\cdot e^{ibt} + e^{-st}\cdot e^{-ibt}\right)dt$$

$$= \frac{1}{2}\int_0^\infty \left(e^{-t(s-ib)} + e^{-t(s+ib)}\right)dt$$

$$= \frac{1}{2}\lim_{a\to\infty}\left[\int_0^a e^{-t(s-ib)}\,dt + \int_0^a e^{-t(s+ib)}\,dt\right].$$

$$= \frac{1}{2}\lim_{a\to\infty}\left\{\left[\frac{e^{-t(s-ib)}}{-(s-ib)}\right]_0^a + \left[\frac{e^{-t(s+ib)}}{-(s+ib)}\right]_0^a\right\}.$$

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$$= \frac{1}{2} \lim_{a\to\infty} \left[\left\{ \frac{1}{ib-s} \left(e^{-a(s-ib)} - e^{0}\right) \right\} + \left\{ \frac{1}{-ib-s} \left(e^{-a(s+ib)} - e^{0}\right) \right\} \right]$$

$$= \frac{1}{2} \left(-\frac{1}{ib-s} - \frac{1}{-ib-s} \right)$$

$$= \frac{1}{2} \left(\frac{1}{s-ib} + \frac{1}{s+ib} \right)$$

$$= \frac{1}{2} \left(\frac{s+ib+s-ib}{(s-ib)(s+ib)} \right)$$

$$= \frac{2s}{2(s+ib)(s-ib)}$$

$$= \frac{s}{s^2 - i^2 b^2}$$

$$= \frac{s}{s^2+b^2}$$

$$\therefore \boxed{\mathcal{L}\{(\cos bt)\} = \frac{s}{s^2+b^2}}$$

$$\mathcal{L}\{\cos 2t\} = \frac{s}{s^2+2^2}$$

$b =$ any constant.

Formulae:

① $\mathcal{L}\{1\} = \frac{1}{s}$; $\mathcal{L}\{2\} = \mathcal{L}\{2\cdot(1)\} = 2\cdot\mathcal{L}\{1\} = 2\cdot 1/s = 2/s$.

$\mathcal{L}\{\text{any constant}\} = \frac{\text{constant}}{s}$ $\Rightarrow \mathcal{L}\{-2\} = \frac{-2}{s}$.

② $\mathcal{L}\{t\} = \frac{1}{s^2} \rightarrow$ Derivation

③ $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \rightarrow \mathcal{L}\{t^3\} = \frac{3!}{s^{3+1}} = \frac{6}{s^4}$

④ $\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \rightarrow \mathcal{L}\{e^{-5t}\} = \frac{1}{s-(-5)} = \frac{1}{s+5}$

⑤ $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$

⑥ $\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$

⑦ $\mathcal{L}\{\sinh at\} = \frac{a}{s^2-a^2}$

} Derivation *

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(8) $\mathcal{L}\{\cosh at\} = \frac{s}{s^2-a^2}\}$ Derivation #-

Type-1

Linearity of Laplace Transform :

$\mathcal{L}\{f(t) \pm g(t)\} = \mathcal{L}\{f(t)\} \pm \mathcal{L}\{g(t)\}$.

$\mathcal{L}\{k f(t)\} = k\,\mathcal{L}\{f(t)\}$.

Eg 1: $\mathcal{L}\{8t^3 - e^{-t}\}$

$= \mathcal{L}\{8t^3\} - \mathcal{L}\{e^{-t}\}$.

$= 8\mathcal{L}\{t^3\} - \mathcal{L}\{e^{-t}\}$.

$= 8 \cdot \frac{3!}{s^{3+1}} - \frac{1}{s-(-1)}$

$= \frac{8 \cdot 3!}{s^4} - \frac{1}{s+1}$ Ans:

Eg-2: $\mathcal{L}\{(t^2+1)^2\}$.

$= \mathcal{L}\{t^4 + 2t^2 + 1\}$

$= \mathcal{L}\{t^4\} + \mathcal{L}\{2t^2\} + \mathcal{L}\{1\}$

$= \frac{4!}{s^{4+1}} + 2 \cdot \frac{2!}{s^{2+1}} + \frac{1}{s}$

$= \frac{4!}{s^5} + 2 \cdot \frac{2!}{s^3} + \frac{1}{s}$. Ans:

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(H.W.) $\mathcal{L}\{\sin^2 at\}$.

$$\boxed{\begin{aligned} 2\sin^2 x &= 1-\cos 2x \\ 2\cos^2 x &= 1+\cos 2x \end{aligned}}$$

Type-2

First translation theorem:

$$\mathcal{L}\{e^{at}.f(t)\} = \mathcal{L}\{f(t)\}_{s\to s-a}$$

Eg-1: $\mathcal{L}\{5e^{3t}.\sin 4t\}$

$= 5\mathcal{L}\{e^{3t}.\sin 4t\}$.

Here, $f(t) = \sin 4t$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{\sin 4t\} = \dfrac{4}{s^2+4^2}$

$\therefore 5\mathcal{L}\{e^{3t}.\sin 4t\} = 5.\dfrac{4}{s^2+4^2}\Big|_{s\to s-3} = 5.\dfrac{4}{(s-3)^2+4^2}$

$= \dfrac{20}{(s-3)^2+16}$ **Ans:**

Eg-2: $\mathcal{L}\{(t+2)^2 e^t\}$

$f(t) = (t+2)^2$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{(t+2)\}^2$

$= \mathcal{L}\{t^2+4t+4\}$

$= \mathcal{L}\{t^2\} + 4\mathcal{L}\{t\} + \mathcal{L}\{4\}$

$= \dfrac{2!}{s^{2+1}} + 4/s^2 + \dfrac{4}{s}$

$= \dfrac{2!}{s^3} + \dfrac{4}{s^2} + \dfrac{4}{s}$

$\therefore \mathcal{L}\{(t+2)^2 e^t\} = \dfrac{2!}{(s-1)^3} + \dfrac{4}{(s-1)^2} + \dfrac{4}{(s-1)}$ **Ans:**

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1. $\mathcal{L}\{e^{at}f(t)\}$
$\Downarrow$
$\mathcal{L}\{f(t)\}\big|_{s\to(s-a)}$

2. linearity
$\mathcal{L}\{t^5\} = \frac{5!}{s^{5+1}}$

Type-3

$$\mathcal{L}\{t^n f(t)\} = ?$$

$$\mathcal{L}\{t^n f(t)\} = \overset{③}{(-1)^n} \overset{②}{\frac{d^n}{ds^n}[F(s)]},$$

where $\underset{\text{1st step}}{F(s)} = \mathcal{L}\{f(t)\}$

1st step $\to F(s)$

2nd step $\to \frac{d^n}{ds^n}[F(s)]$

3rd step $\to$ multiply $(-1)^n$

Ex-1: $\mathcal{L}\{t\underbrace{\sin 2t}_{f(t)}\}$

Soln: Here, $f(t) = \sin 2t$.

$\therefore F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{\sin 2t\} = \frac{2}{s^2+2^2}$

$\therefore \mathcal{L}\{\underset{n=1}{t}\sin 2t\} = (-1)^1 \frac{d^1}{ds^1}\{F(s)\}$

$= -\frac{d}{ds}\left(\frac{2}{s^2+4}\right)$

$= -2\frac{d}{ds}(s^2+4)^{-1}$

$= (-2)(-1)(s^2+4)^{-2}\cdot 2s$

$= \frac{4s}{(s^2+4)^2}$ Ans.

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Last Last 1st

Ex-2: $\mathcal{L}\{e^{4t}\cdot \underbrace{t\sin 2t}\}$. ← Type 2 + Type 3.

Sol^n: $f(t) = t\sin 2t$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{\overset{t^n}{t}\underbrace{\sin 2t}_{g(t)}\}$.

$\therefore G(s) = \mathcal{L}\{\sin 2t\} = \dfrac{2}{s^2+4}$

$$\therefore \mathcal{L}\{t\sin 2t\} = (-1)^1 \frac{d^1}{ds^1}\{G(s)\}$$

$$= -\frac{d}{ds}\left(\frac{2}{s^2+4}\right)$$

$$= -2\frac{d}{ds}(s^2+4)^{-1}$$

$$= (-2)(-1)(s^2+4)^{-2}\cdot 2s$$

$$= \frac{4s}{(s^2+4)^2}$$

$$\therefore \mathcal{L}\{\overset{e^{at}}{\boxed{e^{4t}}}\underbrace{t\sin 2t}_{f(t)}\} = \mathcal{L}\{t\sin 2t\}_{s\to(s-4)}$$

$$= \left.\frac{4s}{(s^2+4)^2}\right|_{s\to(s-4)}$$

$$= \frac{4(s-4)}{\{(s-4)^2+4\}^2}$$

Ans:

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Type-4

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = ?$$

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\,du,$$

where, $F(s) = \mathcal{L}\{f(t)\}$.

$\int 2x\,dx$, $\int 2t\,dt$ } same

$F(s) \xrightarrow{u} F(u)$

power of $\frac{1}{t} = 1$.

1st step $\rightarrow$ $F(s) = \mathcal{L}\{f(t)\}$

2nd step $\rightarrow$ $F(s) \rightarrow F(u)$

3rd step $\rightarrow$ integration

Ex-1: $\mathcal{L}\left\{\frac{\sin 3t}{t}\right\}$

Solⁿ: Here, $f(t) = \sin 3t$

$\therefore F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{\sin 3t\} = \frac{3}{s^2+9}$ $\quad \therefore F(u) = \frac{3}{u^2+9}$

$$\therefore \mathcal{L}\left\{\frac{\sin 3t}{t}\right\} = \int_s^\infty F(u)\,du = \int_s^\infty \frac{3}{u^2+9}\,du$$

$$= \lim_{a\to\infty} \int_s^a \frac{3}{u^2+9}\,du$$

$$= 3 \lim_{a\to\infty} \int_s^a \frac{1}{u^2+9}\,du$$

$$= 3 \lim_{a\to\infty} \int_s^a \frac{du}{u^2+3^2}$$

$$= 3 \lim_{a\to\infty} \left[\frac{1}{3}\tan^{-1}\frac{u}{3}\right]_s^a$$

$$= \lim_{a\to\infty} \left[\tan^{-1}\frac{a}{3} - \tan^{-1}\frac{s}{3}\right] = \tan^{-1}(\infty) - \tan^{-1}\left(\frac{s}{3}\right).$$

$$= \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{3}\right).$$ Ans:

$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$

$\lim_{x\to 5}(t^2) = t^2$

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← Type 2 + Type 4.

Ex-2: $\mathcal{L}\left\{e^{4t}\,\frac{\sin 3t}{t}\right\} = ?$

Soln: $\mathcal{L}\left\{\boxed{e^{4t}}\,\underbrace{\frac{\sin 3t}{t}}_{f(t)}\right\} = \mathcal{L}\left\{\frac{\sin 3t}{t}\right\}_{s\to(s-4)}.$

$\therefore \mathcal{L}\left\{\frac{\sin 3t}{t}\right\} = \int_s^\infty G(u)\,du$, let $g(t) = \sin 3t$

Here, $G(s) = \mathcal{L}\{g(t)\} = \mathcal{L}\{\sin 3t\}. = \frac{3}{s^2+9}$

$\therefore G(u) = \frac{3}{u^2+9}$

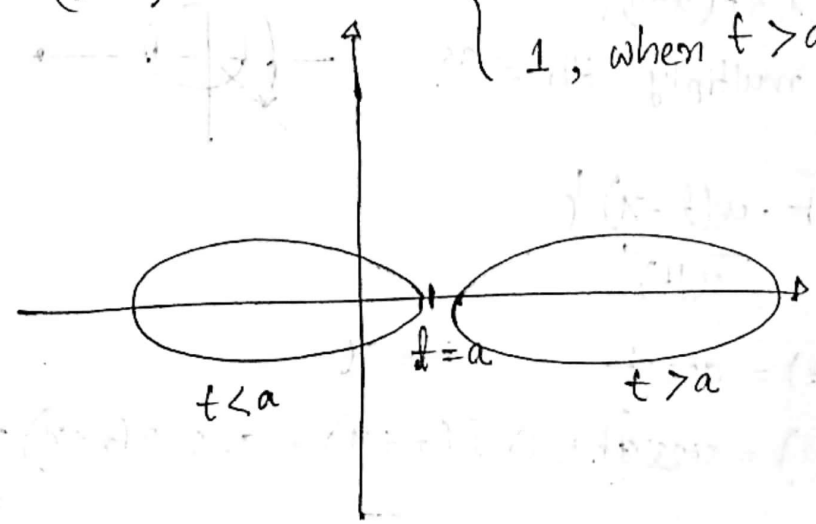
$\therefore \mathcal{L}\left\{\frac{\sin 3t}{t}\right\} = \int_s^\infty G(u)\,du = \int_s^\infty \frac{3}{u^2+9}\,du = \frac{\pi}{2} - \tan^{-1}(s/3).$

$\therefore \mathcal{L}\left\{e^{4t}\frac{\sin 3t}{t}\right\} = \frac{\pi}{2} - \tan^{-1}(s/3)\Big|_{s\to(s-4)}$

$= \frac{\pi}{2} - \tan^{-1}\left(\frac{s-4}{3}\right).$ **Ans:**

Unit step function:

Defn: $u(t-a) = u_a(t) = \begin{cases} 0, & \text{when } t<a \\ 1, & \text{when } t>a. \end{cases}$



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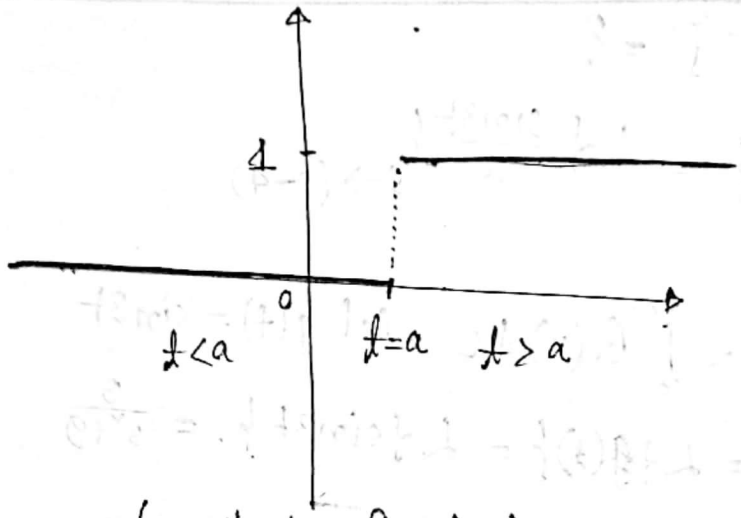


(unit step function)

$$\boxed{\mathcal{L}\{u_a(t)\} = \frac{e^{-as}}{s}} \quad / \quad \boxed{\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}}$$

Type-5

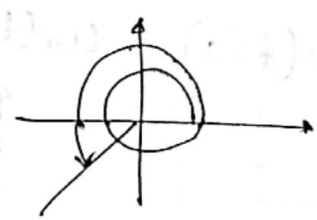
2nd Translation Theorem

$$\mathcal{L}\{f(t)\,u(t-a)\} = \mathcal{L}\{f(t+a)\}\,e^{-as}$$

1st step $\rightarrow f(t) \rightarrow f(t+a)$

2nd step $\rightarrow \mathcal{L}\{f(t+a)\}$

3rd step $\rightarrow$ multiply with e^{-as}.



Ex-1: $\mathcal{L}\{\underbrace{\cos 3t}_{f(t)} \cdot \underbrace{u(t-\pi)}_{u_a(t)}\}$

Solⁿ: Here, $f(t) = \cos 3t$; $a = \pi$

$\therefore f(t+a) = \cancel{\cos(3t+3\pi)}\ f(t+\pi) = \cos 3(t+\pi) = \cos(3t+3\pi)$

$= -\cos 3t$

$\therefore \mathcal{L}\{f(t+a)\} = \mathcal{L}\{f(t+\pi)\} = \mathcal{L}\{-\cos 3t\} = -\frac{s}{s^2+9}$

$\therefore \mathcal{L}\{\cos 3t \cdot u(t-\pi)\} = \mathcal{L}\{f(t+\pi)\} e^{-\pi s}$

$= -\frac{s}{s^2+9} \cdot e^{-\pi s}$. **Ans:**

(H.W.) $\boxed{\mathcal{L}\{\cos 3t \cdot u(t-\pi) \cdot e^{2t}\} = ?}$

Type-6

Laplace of piece-wise function :

$$f(t) = \begin{cases} \sin t, & 0 \le t < \pi \\ \cos t, & \pi \le t < 2\pi \\ 0, & t \ge 2\pi. \end{cases}$$ piecewise function.

(1) $\boxed{\text{convert to unitstepfunction.}}$ → (2) Then do like Type 5.

(*) $u(t-0) = 1$
$u_0(t)$

(*) $u(t-\infty) = 0$
$u_\infty(t)$

$f(t) =$ 1st functional value $\left(u(t - \text{lower limit}) - u(t - \text{upper limit})\right)$
$+ 2^{nd}$ - - - - -
$+ 3^{rd}$ - - - - -

Ex-1: $f(t) = \begin{cases} 0 & ; 0 \le t < \pi \\ \sin t & ; \pi \le t < 2\pi \\ e^{3t} & ; t \ge 2\pi. \end{cases}$

$2\pi \le t < \infty$

Find $\mathcal{L}\{f(t)\}$.

Soln: $f(t) = 0\left(u(t-0) - u(t-\pi)\right) +$
$\sin t\left(u(t-\pi) - u(t-2\pi)\right) +$
$e^{3t}\left(u(t-2\pi) - u(t-\infty)\right)$.

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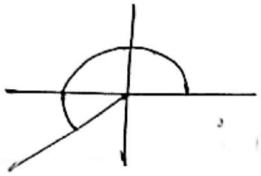
$\therefore f(t) = \sin t \cdot u(t-\pi) - \sin t \cdot u(t-2\pi) + e^{3t} \cdot u(t-2\pi)$.

$$\boxed{\mathcal{L}\{f(t)\,u(t-a)\} = \mathcal{L}\{f(t+a)\}e^{-as}}$$

$\therefore \mathcal{L}\{\sin t \cdot u(t-\pi)\} = \mathcal{L}\{\sin(t+\pi)\}e^{-\pi s}$

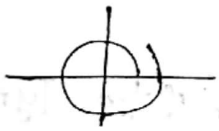


$= \mathcal{L}\{-\sin t\}e^{-\pi s}$

$= -\dfrac{1}{s^2+1}e^{-\pi s}$.

Now,

$\therefore \mathcal{L}\{\sin t \cdot u(t-2\pi)\} = \mathcal{L}\{\sin(t+2\pi)\}e^{-2\pi s}$



$= \mathcal{L}\{\sin t\}e^{-2\pi s}$

$= \dfrac{1}{s^2+1}e^{-2\pi s}$.

And,

$\mathcal{L}\{e^{3t} \cdot u(t-2\pi)\} = \cancel{\mathcal{L}\{e^{3t}\}e^{-2\pi s}}\ \mathcal{L}\{\underbrace{e^{3(t+2\pi)}}_{f(t+a)}\}e^{-2\pi s}$

$= \cancel{\dfrac{1}{s-3}e^{-2\pi s}}$

$\boxed{\mathcal{L}/\Lambda}$

Laplace

$1+2\pi$

$\frac{1}{s-1} \cdot \frac{2\pi}{s-1}$

$= \mathcal{L}\{e^{3t} \cdot e^{6\pi}\}e^{-2\pi s}$

$= e^{6\pi} \cdot \mathcal{L}\{e^{3t}\}e^{-2\pi s}$

$= e^{6\pi} \cdot \dfrac{1}{s-3} \cdot e^{-2\pi s}$

$= \dfrac{1}{s-3} \cdot e^{6\pi} \cdot e^{-2\pi s}$

$\therefore \mathcal{L}\{f(t)\} = -\dfrac{1}{s^2+1}e^{-\pi s} - \dfrac{1}{s^2+1}e^{-2\pi s} + e^{6\pi} \cdot \dfrac{1}{s-3} \cdot e^{-2\pi s}$ Ans:

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